

OM RAHEJA

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Education

University of Illinois Urbana-Champaign (UIUC)

Expected May 2029

Bachelor of Science in Computer Engineering

Urbana, IL

Technical Skills

Languages: Python, JavaScript, SQL, Go, C, x86 Assembly, GDScript, HTML/CSS

Frameworks & Libraries: Next.js, React, Node.js, Express, PostgreSQL, Groq API, Gemini API, TensorFlow.js, Axios

AI & Systems: Agentic AI, Vision-Language Models (VLMs), OCR, Structured Data Extraction, Multi-Agent Systems, OpenClaw, AutoFlow, QEMU, Linux Internals

Tools & Analytics: Git, GitHub, Google Analytics, beehiiv, Godot Engine (2D/3D), OpenCAD, OnShape, Blender

Work Experience

codeabode | Python, Java, JJS, Cloudflare Workers, Next.js

April 2024 – Present

Chief Executive Officer & Lead Developer

Plainsboro, NJ

- Coordinated a 7-person team to deliver 70+ hours of instruction to 100+ students, generating \$2,000 in revenue
- Architected a Python AI pipeline to dynamically generate personalized programming curricula based on real-time student performance metrics
- Designed an interactive 3D game with three.js to visually simulate core programming paradigms for students

AIOS – Rutgers University | LaTeX, Python, LiteLLM, AgentBench

May 2024 – Dec. 2025

Research Assistant

New Brunswick, NJ

- Co-authored AutoFlow, an automatic agentic workflow generator optimizing LLM task execution and multi-agent coordination
- Addressed 50+ students & PhDs on agentic infrastructure architectures at the AIOS Hackathon
- Specified comprehensive open-source documentation covering framework installation, system usage, and contributor guidelines

aquarc | Selenium, Go, React, PostgreSQL, VectorDB, Google Analytics

June 2024 – July 2025

Chief Executive Officer & Lead Architect

Remote

- Innovated an enterprise SAT prep platform scaling to 8,000+ questions using Go, React, and PostgreSQL
- Delegated tasks within a distributed remote team of 10+ developers from product conception through agile deployment cycles
- Drove growth to 1,000+ active users by leveraging Google Analytics and launching a targeted beehiiv newsletter

Project Experience

SwimIQ (swim-iq.vercel.app) | React, TensorFlow.js, Express, Groq Llama 4 Scout

July 2026

- Engineered a real-time computer vision web app using TensorFlow.js and MoveNet Thunder for in-browser pose estimation, tracking 17 keypoints at 30 FPS
- Customized a hybrid edge/cloud architecture utilizing Groq's Llama 4 Scout vision model to generate coaching feedback
- Incorporated a custom kinematic engine in React to identify stroke types and phase despite real-world water glare and splashing

MeetTimeTracker | Python, Gemini API, OCR, JSON

Jan. 2026

- Built an AI agent using Gemini's OCR to extract and parse unstructured physical swim meet results into a centralized database
- Formulated custom prompt algorithms to resolve complex data row alignment errors and overlapping text in scanned medley relay results
- Automated the conversion of image data into structured JSON payloads, eliminating manual data entry

omOS | C, x86 Assembly, QEMU

Feb. 2024

- Wrote a standalone C standard library and a VGA graphics driver integrated with a hardware interrupt keyboard handler
- Utilized QEMU for system emulation, kernel debugging, and low-level memory mapping

Leadership & Advocacy

Legislative Advocacy & AIxEducation | National Finalist & Co-Author

Sep. 2024 – Nov. 2024

- Won 2nd Place Nationally for AI research simulating collaborative Montessori schools, and co-authored the Comprehensive Mental Health Support Act, securing state senate co-sponsorship

New Jersey Privacy Movement | Founder

Jan. 2024 – Mar. 2024

- Authored a public petition for user data rights and collaborated with NJ State Senator Turner to draft a formal privacy bill

FBLA | 7th Place Regional Champion

Sep. 2022 – June 2026

- Ranked 7th in Computer Problem Solving for Central NJ; raised \$150+ and collected 100+ shoes for charity drives